

Do Lifestyle and Work Habits Predict Longevity? A Machine Learning Approach

Katie Hur, Dylan Miner

Code: <https://github.com/dylanminer27/cpsc3810-longevity.git>

CPSC 3810: Introduction to Machine Learning

Instructor: Alex Wong

Final Project Report - May 6, 2026

Introduction

“Work-life balance,” first coined in 1986, has received substantial attention from public health research in recent years, what the WHO and APA call the stress epidemic [1]. Research on the health effects of stress and burnout among workers has even fueled initiatives for workers reasserting their personal time [2]. Allocation of one’s personal time across work, rest, sleep, and exercise has been hypothesized to influence a person's expected lifespan. Other factors include gender and occupation.

In this final project, we present the relationship between an individual’s lifestyle and lifespan as a regression problem. Leveraging various machine-learning techniques, we aim to build a regression model that can accurately predict one’s expected longevity from gender, occupation, and daily time allocation using the *Work-Life Balance and Longevity Dataset* [3]. From this model, we hope to attain an improved understanding of which lifestyle factors are most closely correlated with lifespan.

We approached this problem through:

1. **Linear regression** (supervised) to predict age at death directly from lifestyle and demographic features.
1. **Multinomial logistic regression** (supervised) to classify individuals into three longevity tiers, Short, Medium, or Long, formed from training-set tertile cutoffs of age at death.
2. **K-Means clustering** (unsupervised) to discover natural lifestyle archetypes among the 10,000 individuals in the dataset, then examine each cluster's mean age at death.

Each model is implemented from scratch and compared to the corresponding scikit-learn baseline. One challenge we faced was modest predictive power of the available features, which capped how high our metrics could reach.

Data

We used a publicly available, synthetic dataset downloaded from Kaggle, called [*Work-Life Balance and Longevity Dataset*](#), which contains records for 10,000 individuals. It contains information in tabular form on 10,000 individuals, each described by seven features: their gender, occupation type (across fourteen categories), daily work hours, daily rest hours, daily sleep hours, and daily exercise hours. The target variable is age at death.

Column Descriptions		
Column Name	Type	Description
id	Integer	Unique identifier for each individual
gender	Text	Gender of the individual (Male or Female)
occupation_type	Text	Type of occupation (14 categories: Healthcare Worker, Teacher, Engineer, Artist, Manual Laborer, Office Worker, Entrepreneur, Retail Worker, Driver, Scientist, Freelancer, Manager, Technician, Consultant)
avg_work_hours_per_day	Float	Average hours spent working per day (0-24 hours). Includes outliers with extreme work hours up to 24 hours.
avg_rest_hours_per_day	Float	Average hours spent resting per day (0-24 hours). Includes outliers with extreme rest hours up to 24 hours.
avg_sleep_hours_per_day	Float	Average hours spent sleeping per day (0-24 hours)
avg_exercise_hours_per_day	Float	Average hours spent exercising per day (0-4 hours)
age_at_death	Integer	Age at which the individual died, representing longevity (25-100 years)

Figure 1: *Work-Life Balance and Longevity Dataset* schema [3]

The target spans 25–100 with a mean of 79.85 and standard deviation of 12.03. This dataset is well-suited for our regression problem. It is sufficiently large to support a meaningful train-validation-test split. Additionally, most of the features and the target are continuous real values. Categorical features (gender and occupation type) were one-hot encoded in the data preparation process prior to training, producing a binary `gender_Male` feature and fourteen binary occupation indicator features (e.g., `occupation_type_Consultant`).

It is important to acknowledge that the dataset is synthetic, which may not capture the variance, confounding variables (e.g. socioeconomic status, diet, access to healthcare), or distributional complexity present in real-world data. However, the dataset does include outliers

that represent extreme work or rest patterns, which nuances our exploration with non-linear relationships and diminishing returns in health outcomes [3]. Nevertheless, results should be interpreted with this limitation in mind and are not necessarily generalizable to real populations.

Methodology

Our methodology incorporates three methods from class: linear regression, logistic regression, and clustering.

To directly predict age at death, we implemented a linear regression model from scratch to be compared against a scikit-learn baseline. Our custom implementation supports three solvers: the normal equation solver used the closed-form formula $XX^T w = Xy$ with `np.linalg.solve`; the pseudoinverse solver computed an SVD solution with `np.linalg.pinv`; and the gradient descent solver used numerical approximation. The gradient descent (GD) approach conducted an optimizer sweep across four configurations: GD, stochastic GD, momentum GD, and stochastic momentum GD. We compared convergence speed and final loss (in MSE) across variants.

The objective of the logistic regression approach was classification, for which we categorized age at death into Short, Medium, and Long tiers. We implemented a multinomial softmax classifier, also to be compared against a scikit-learn baseline, trained on momentum GD and optimizing for cross-entropy loss.

While the first two approaches leveraged supervised algorithms, our K-means clustering approach was unsupervised, identifying patterns and naturally formed groups. This method only used the four continuous features (work, rest, sleep, exercise hours). Sweeping k from 2 to 7, we used an elbow plot of inertia to select $k = 4$. For each resulting cluster, we computed the mean age at death to examine any correlation with longevity.

Implementation Details

Data preparation

First, the `src/data.py` script standardized the raw data to zero-mean, unit-variance values and formatted it into a clean dataset of features and ground truth (age at death). The average work/rest/sleep/exercise hours per day features were already formatted as continuous, real

numbers. The processed data was then split into training, validation, and test sets. A bias term was also prepended to each feature vector in the model's `fit` method.

Linear regression

The `src/train.py` script executes the training phase of two models: Scikit-learn's built-in Linear Regression model as a baseline and our own custom regression model defined in `src/model.py`. Two closed-form solvers were implemented: the normal equation and pseudoinverse, neither of which required hyperparameters. The `optimizer_sweep` method compared final loss *and* convergence speed among four different optimizer configurations as follows:

Optimizer type	alpha	eta decay factor	beta	Batch size
<code>gradient_descent</code>	1e-2	0.0	0.0	None (full)
<code>stochastic_gradient_descent</code>	1e-2	0.5	0.0	256
<code>Momentum_gradient_descent</code>	1e-2	0.0	0.9	None (full)
<code>momentum_stochastic_gradient_descent</code>	1e-2	0.5	0.9	256

Figure 2: Our custom linear regression model's optimizers and hyperparameters

Stochastic-based algorithms incorporated an eta (decay factor) and batch size hyperparameter, used to shrink learning rate and adjust training data subset size respectively. Momentum-based algorithms incorporated a beta (momentum coefficient) hyperparameter, used to accumulate a weighted average of past gradients. Learning rate followed polynomial decay. Mini-batches were sampled without replacement.

Logistic regression

Our multinomial softmax classifier was trained with momentum SGD (using the same corresponding hyperparameters in Figure 2).

K-means clustering

Clustering was performed exclusively on the four continuous lifestyle features. K was swept from 2 to 7, and we selected $k = 4$ via elbow analysis of inertia. Then, the mean age at death was computed for each cluster.

Results

Linear regression results:

The closed-form solvers of our linear regression reproduce the scikit-learn baseline almost exactly, and the gradient-descent versions converge to within $\sim 2\%$ of the closed-form optimum.

Model	Train MSE	Train R^2	Val MSE	Val R^2	Test MSE	Test R^2
Sklearn baseline	103.69	0.279	107.37	0.277	104.78	0.291
Ours — normal equation	103.69	0.279	107.43	0.276	104.86	0.290
Ours — pseudoinverse	103.69	0.279	107.37	0.277	104.78	0.291
Ours — GD	104.39	0.274	108.57	0.269	106.85	0.277
Ours — momentum GD	104.38	0.274	108.57	0.269	106.84	0.277
Ours — SGD	128.90	0.103	131.95	0.111	131.89	0.107
Ours — momentum SGD	104.41	0.274	108.42	0.270	106.84	0.277

Figure 3: Our linear regression model vs Scikit-learn baseline

The modest test R^2 of 0.29 reflects an upper bound on what a linear model can extract from these six features. Roughly 71% of the variance in age at death is unexplained by the lifestyle inputs alone, consistent with the dataset being synthetic and noisy rather than a deterministic function of work, rest, sleep, and exercise.

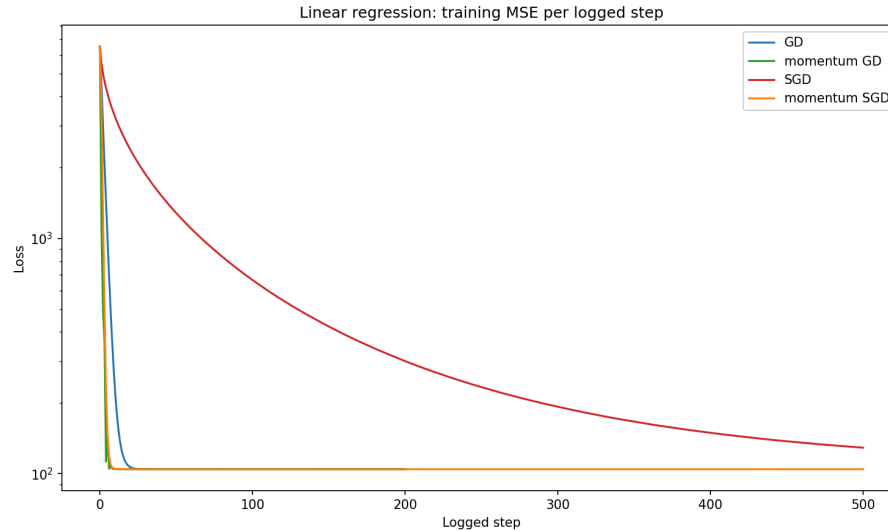


Figure 4: Training loss in our model compared across four optimizers

Based on results in Figure 4 above, momentum GD converged fastest while SGD converged slowest. GD and momentum GD achieved similar performance, landing at the same loss, but adding the momentum coefficient for full-batch GD slightly improved performance. On the other hand, the large gap between SGD and momentum SGD indicates that the momentum coefficient plays a much more significant role in the stochastic case. This follows from the fact that SGD’s path is increasingly noisier as it converges.

Logistic regression and classification results:

Our multinomial logistic regression matches the scikit-learn baseline to within 0.001 on test accuracy, again confirming that our implementation captures the same hypothesis class.

Model	Train acc.	Val acc.	Test acc.	Test F1 (macro)
Sklearn baseline	0.512	0.506	0.498	0.441
Ours — momentum SGD	0.511	0.502	0.499	0.430

Figure 5: Our logistic regression model vs Scikit-learn baseline

A 50% test accuracy reflects real signal, but the per-class breakdown reveals a structural failure of the linear models on this task:

Class (custom model, test)	Precision	Recall	F1
Short	0.508	0.650	0.571
Medium	0.355	0.070	0.116
Long	0.509	0.740	0.603

Figure 6: Per-class breakdown of precision, recall, and F1 scores for our model on test set. The Medium class achieves only 7% recall which means both models predict Short or Long and almost never choose Medium. This is visible in both confusion matrices (Figures 7 and 8). The center column is sparse for both our model and the scikit-learn model. This means that the phenomenon is not an error in our implementation, but rather a property of the problem: with the available linear features, the boundary between “medium lifespan” (ages 77-85) and the short and long category is essentially invisible.

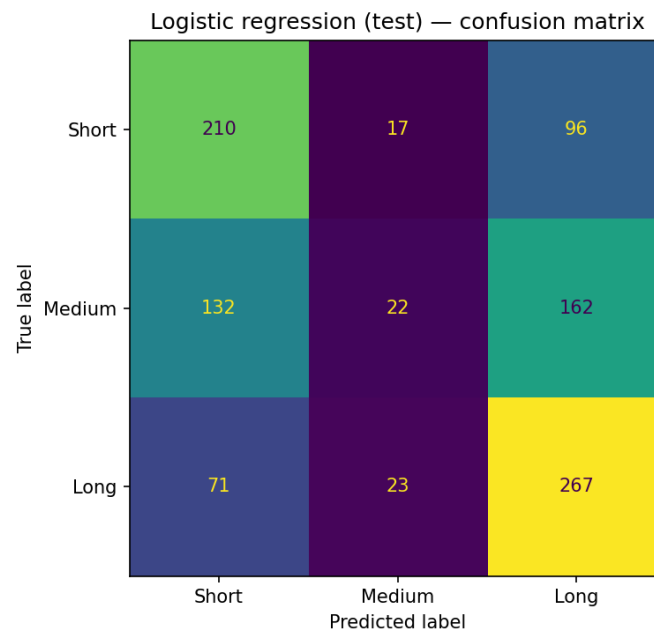


Figure 7: Confusion matrix for our model on test set

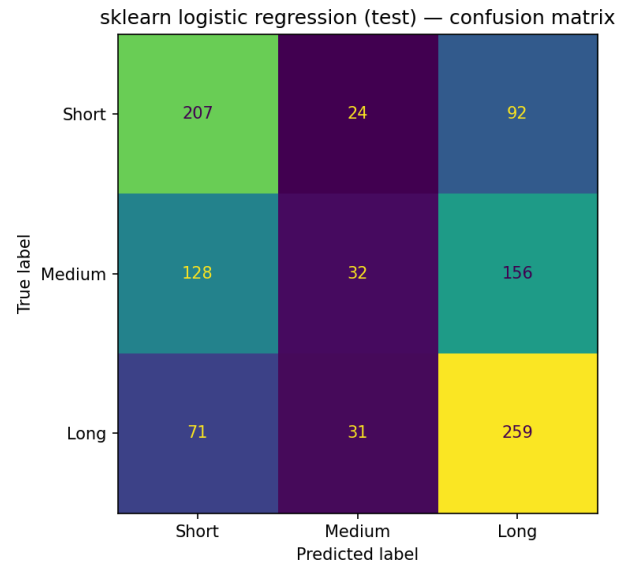


Figure 8: Confusion matrix for Scikit-learn model on test set

Clustering results:

Inertia decreases smoothly from $k=2$ to $k=7$ without a sharp elbow, as seen in Figure 9. But, at $k=4$, K-Means cleanly isolates a small “extreme overwork” while the remaining three clusters represent more balanced lifestyle types.

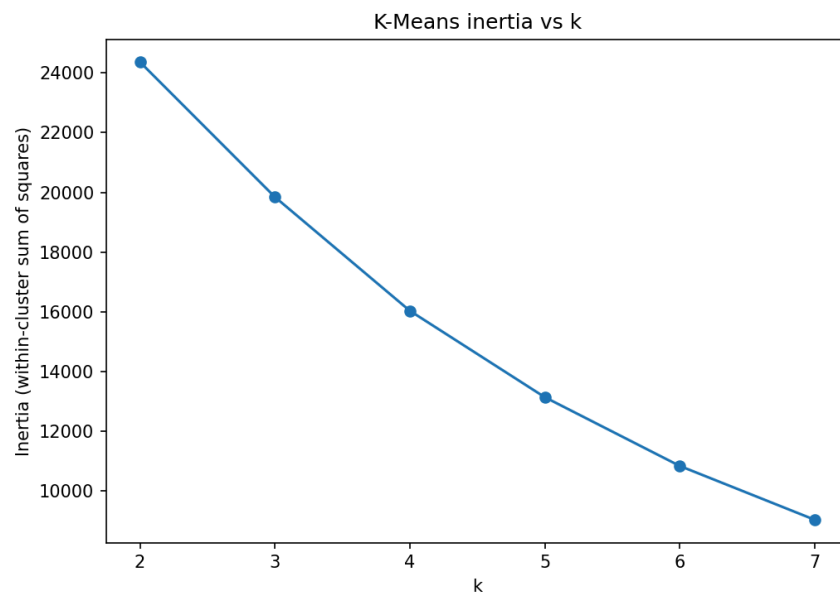


Figure 9: K-means elbow plot

The four clusters at $k = 4$ are summarized below. Lifestyle values are reported in standard deviation units, where 0 is the population mean for each feature. Cluster 0 is the headline finding. It contains only 168 of the 10,000 individuals, but its members have lifestyle profiles up to 4 standard deviations above the mean for work hours, as well as below-average rest, sleep, and exercise. Their mean age at death is 35.5, which is around 45 years younger than the more balanced lifestyle clusters. The bar chart in Figure 12 shows this gap clearly.

Cluster	n	Mean age at death	Work	Rest	Sleep	Exercise
0	168	35.5	+4.00	-1.30	-2.74	-1.53
1	2673	77.4	+0.05	-0.19	+0.55	-0.80
2	2046	84.4	-0.71	+1.19	-0.67	-0.25
3	3113	81.5	+0.20	-0.55	+0.12	+0.93

Figure 10: K-means cluster summary at $k = 4$

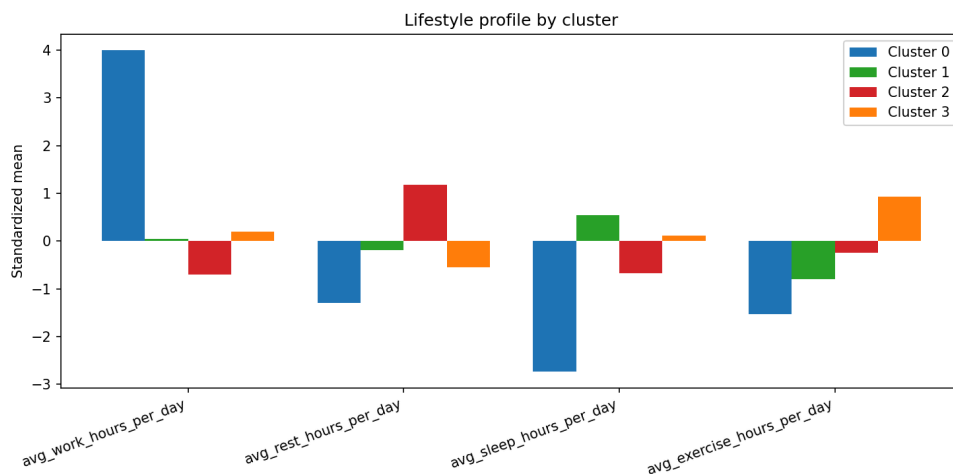


Figure 11: Lifestyle profile by cluster

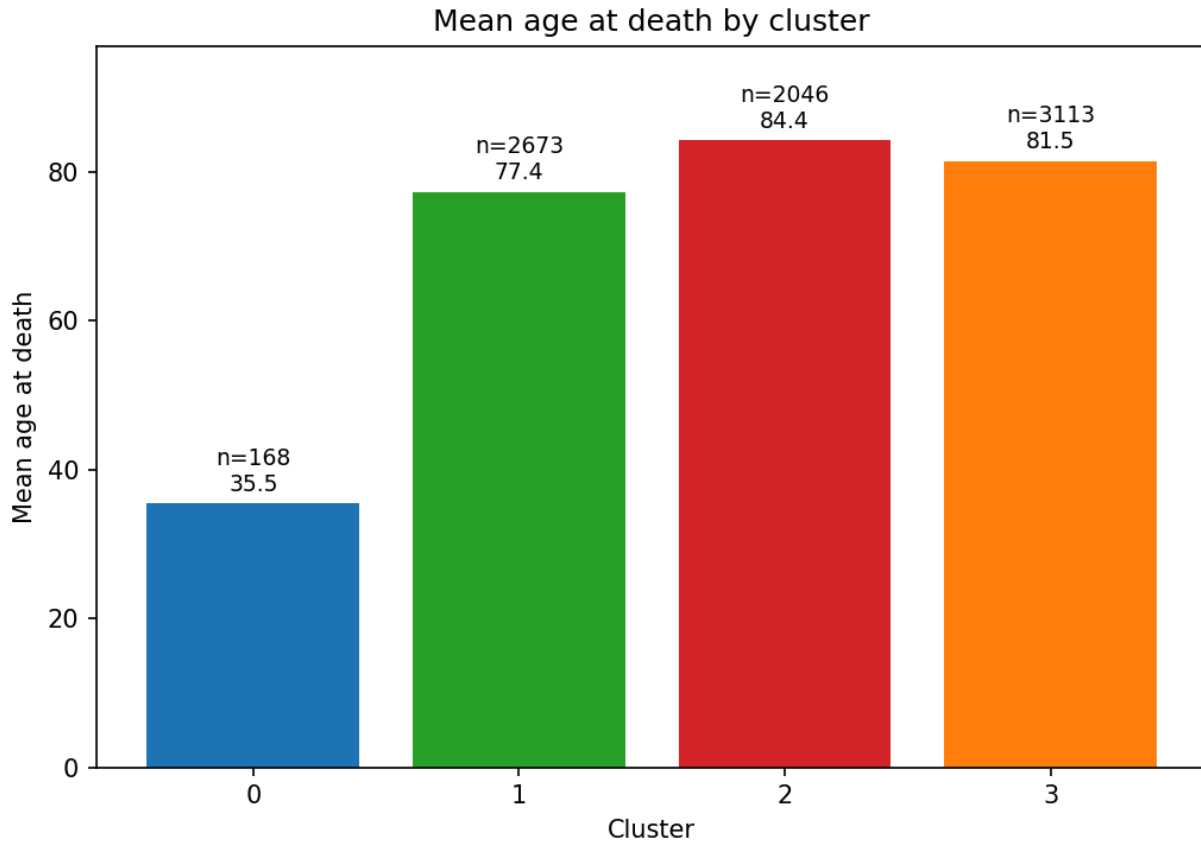


Figure 12: Mean age at death by cluster

Among the three balanced lifestyle clusters, Cluster 2 — characterized by below-average work hours and the highest rest hours of any group — has the longest mean lifespan (84.4 years). Cluster 3, the high-exercise group, lives almost as long (81.5). Cluster 1, with average work and the highest sleep but lowest exercise, lives shortest of the three (77.4). The ordering across the three balanced clusters is modest but consistent with a broader pattern: more rest and exercise correlate with longer lifespans, whereas overwork at the extreme is associated with dramatically shorter ones.

Summary of findings:

The supervised models confirm that a linear hypothesis can only modestly predict age at death from the available features: the test R^2 ceiling of ~ 0.29 and test classification accuracy of $\sim 50\%$ likely reflect the upper bound of what is extractable. Both custom implementations match scikit-learn to within rounding, demonstrating that the implementations are correct, and the

side-by-side optimizer comparison reproduces the textbook result that momentum is essential for stochastic mini-batches.

The unsupervised analysis surfaces the most actionable insight: a small subgroup (1.7% of the data) defined by extreme overwork accounts for an enormous gap in mean longevity. Within the 98% of individuals who do not work to that extreme, lifestyle differences correlate with much smaller (~7-year) differences in mean age at death. From a methodological standpoint this suggests that for this dataset, clustering surfaces structure that linear regression cannot. The non-linear "extreme overwork" effect is averaged away when treated as a continuous predictor in a linear model.

References

[1] Stress, the “health epidemic of the 21st century.” (2019, May 1). HCA Healthcare Today.

<https://hcahealthcaretoday.com/2019/04/30/stress-the-health-epidemic-of-the-21st-century/>

[2] Patricia Duchene. (2019, April 29). Beyond Work-Life Balance. Forbes.

<https://www.forbes.com/sites/patriciaduchene/2019/04/29/beyond-work-life-balance/>

[3] Oluwatosin Adewale & Tonic Fabricate. (n.d.). Work-Life Balance and Longevity Dataset

[Dataset]. Kaggle. Retrieved March 5, 2026, from

<https://www.kaggle.com/datasets/oluwatosinadewale/quality-of-life-data>